

# BENCHMARK EVALUATION: CONVENTIONAL vs. UPGRADED EPS MORTAR

Comprehensive 2x2 Performance Matrix (Experimental Replicates with Standard Deviations)

| Performance Parameter                     | Conventional EPS<br>(20% Replacement) | Upgraded EPS<br>(20% Replacement) | Conventional EPS<br>(40% Replacement) | Upgraded EPS<br>(40% Replacement) |
|-------------------------------------------|---------------------------------------|-----------------------------------|---------------------------------------|-----------------------------------|
| 28-day Oven-Dry Density (kg/m³)           | 2011.0 ± 18.5                         | 1969.0 ± 17.0 (Lightweight)       | 1748.0 ± 15.0                         | 1734.0 ± 14.5 (Lightweight)       |
| Compressive Strength f <sub>c</sub> (MPa) | 10.15 ± 0.65                          | 15.62 ± 0.85 (+53.9% ▲)           | 4.20 ± 0.35                           | 5.21 - 7.15 (+70.2% ▲)            |
| Flexural Strength f <sub>r</sub> (MPa)    | 2.10 ± 0.15                           | 3.59 ± 0.20 (+71.0% ▲)            | 0.95 ± 0.08                           | 1.80 - 1.85 (+94.7% ▲)            |
| Fracture Toughness (mJ/mm³)               | 12.58 ± 0.85                          | 18.24 ± 1.20 (+45.0% ▲)           | 1.87 ± 0.15                           | 3.84 ± 0.25 (+105.3% ▲)           |
| Apparent Porosity (%)                     | 7.65 ± 0.25%                          | 6.48 ± 0.24% (Refined)            | 8.03 ± 0.30%                          | 6.94 ± 0.26% (Refined)            |
| Water Absorption (%)                      | 3.82 ± 0.15%                          | 3.34 ± 0.14% (Reduced)            | 4.50 ± 0.18%                          | 3.52 ± 0.15% (Reduced)            |
| ITZ Ca/Si Ratio (SEM-EDS)                 | 23.50 ± 2.40 (CH)                     | 6.80 ± 0.55 (C-S-H ▼)             | 24.85 ± 2.70 (CH)                     | 5.65 ± 0.39 (-77.3% C-S-H ▼)      |
| Strength-to-Cost (MPa/kLKR)               | 0.275 ± 0.020                         | 0.323 ± 0.025 (+17.5% ▲)          | 0.098 ± 0.009                         | 0.131 ± 0.011 (+33.7% ▲)          |
| Embodied Carbon (kg CO2/MPa)              | 53.47 ± 2.80                          | 32.29 ± 1.60 (-39.6% ▼)           | 130.02 ± 6.50                         | 95.66 ± 4.50 (-26.4% ▼)           |